

Assignment 4

1. The elliptical integral of the first kind is defined as

$$F(x, k) \equiv \int_0^x \frac{dt}{\sqrt{1 - k^2 \sin^2 t}}, \quad 0 \leq k < 1.$$

Consider a pendulum of mass m and length l displaced by an angle θ from its equilibrium position. By conservation of energy, the angular position $\theta(t)$ of the mass satisfied the following ordinary differential equation:

$$\frac{1}{2}l \left[\frac{d\theta}{dt} \right]^2 - g \cos \theta(t) = -g \cos \theta_0,$$

where θ_0 is the maximum angular displacement. At $t = 0$, we choose as initial conditions $\theta(0) = 0$ and $\left. \frac{d\theta(t)}{dt} \right|_{t=0} > 0$.

(a) Show that the period of oscillation is given by

$$T = 4\sqrt{\frac{l}{g}} F\left(\frac{\pi}{2}, \sin \frac{\theta_0}{2}\right).$$

(b) Hence, evaluate the period of oscillation as powers of $\sin^2(\theta_0/2)$ up to the $\sin^4(\theta_0/2)$ term.

Answer:

(a)

By solving the given differential equation, we derive following integral

$$\int_0^{T/4} dt = \int_0^{\theta_0} \sqrt{\frac{l}{2}} \frac{d\theta}{\sqrt{g \cos \theta - g \cos \theta_0}} \quad (1)$$

The left hand side equals $T/4$. So the period is

$$T = 4\sqrt{\frac{l}{2g}} \int_0^{\theta_0} \frac{d\theta}{\sqrt{\cos \theta - \cos \theta_0}} = 4\sqrt{\frac{l}{2g}} \int_0^{\theta_0} \frac{d\theta}{\sqrt{2 \sin^2 \frac{\theta_0}{2} - 2 \sin^2 \frac{\theta}{2}}} \quad (2)$$

Suppose $\sin \frac{\theta}{2} = \sin \frac{\theta_0}{2} \sin \phi$, as results,

$$d\theta = \frac{2 \sin \frac{\theta_0}{2} \cos \phi \, d\phi}{\sqrt{1 - \sin^2 \frac{\theta_0}{2} \sin^2 \phi}} \quad (3)$$

Now the integral is

$$\begin{aligned} 4\sqrt{\frac{l}{2g}} \int_0^{\theta_0} \frac{d\theta}{\sqrt{2 \sin^2 \frac{\theta_0}{2} - 2 \sin^2 \frac{\theta}{2}}} &= 4\sqrt{\frac{l}{2g}} \int_0^{\pi/2} \frac{\frac{2 \sin \frac{\theta_0}{2} \cos \phi \, d\phi}{\sqrt{1 - \sin^2 \frac{\theta_0}{2} \sin^2 \phi}}}{\sqrt{2 \sin^2 \frac{\theta_0}{2} - 2 \sin^2 \frac{\theta_0}{2} \sin^2 \phi}} \\ &= 4\sqrt{\frac{l}{g}} \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - \sin^2 \frac{\theta_0}{2} \sin^2 \phi}} = 4\sqrt{\frac{l}{g}} F\left(\frac{\pi}{2}, \sin \frac{\theta_0}{2}\right) \end{aligned} \quad (4)$$

Q.E.D.

(b)

$$\begin{aligned} T &\approx 4\sqrt{\frac{l}{g}} \int_0^{\pi/2} \left(1 + \frac{k^2}{2} \sin^2 \phi + \frac{3k^4}{8} \sin^4 \phi + O(k^6) \right) d\phi \\ &= 4\sqrt{\frac{l}{g}} \left(\frac{\pi}{2} + \frac{\pi}{8} \sin^2 \theta_0 + \frac{9\pi}{128} \sin^4 \theta_0 + O(\sin^6 \theta_0) \right) \end{aligned} \quad (5)$$

2. A diatomic molecule may, to a good approximation, be treated as vibrating in one-dimensional simple harmonic motion with angular frequency ω . In statistical mechanics, the average vibrational energy of a diatomic molecule is given by

$$\langle E \rangle = \frac{\sum_{n=0}^{\infty} n\hbar\omega \exp(-n\hbar\omega/kT)}{\sum_{n=0}^{\infty} \exp(-n\hbar\omega/kT)}$$

Where $\hbar = h/2\pi$ is the reduced Planck constant, k is the Boltzmann constant and T is the temperature.

(a) Find an explicit expression for the average vibrational energy of the diatomic molecule.

(b) The molar vibrational heat capacity is defined as

$$C \equiv N_A \frac{d\langle E \rangle}{dT},$$

Where N_A is the Avogadro number. Find an exact expression of the molar heat capacity of the diatomic molecule. Hence, obtain expression for the high and low temperature limits of the molar vibration heat capacity of diatomic molecule.

Answer:

(a)

Suppose $\sum_{n=0}^{\infty} n\hbar\omega \exp(-n\hbar\omega/kT) = S_n$, $\sum_{n=0}^{\infty} \exp(-n\hbar\omega/kT) = T_n$

$$\lim_{n \rightarrow \infty} S_n = \frac{e^{-\hbar\omega/kT}}{(1 - e^{-\hbar\omega/kT})^2}, \quad \lim_{n \rightarrow \infty} T_n = \frac{1}{1 - e^{-\hbar\omega/kT}} \quad (6)$$

As the result, the explicit expression for the average vibrational energy of the diatomic molecule is

$$\langle E \rangle = \frac{\hbar\omega}{e^{\hbar\omega/kT} - 1} \quad (7)$$

(b)

The exact expression of the molar heat capacity of the diatomic molecule is

$$C \equiv N_A \frac{d\langle E \rangle}{dT} = \frac{N_A \hbar^2 \omega^2}{kT^2 (e^{\hbar\omega/kT} - 1)^2} \quad (8)$$

high temperature limit:

$$\lim_{\hbar\omega/kT \rightarrow 0} \frac{N_A \hbar^2 \omega^2}{kT^2 (e^{\hbar\omega/kT} - 1)^2} = N_A k \quad (9)$$

low temperature limit:

$$\lim_{\hbar\omega/kT \rightarrow \infty} \frac{N_A \hbar^2 \omega^2}{kT^2 (e^{\hbar\omega/kT} - 1)^2} = 0 \quad (10)$$

3. Consider an arrangement of N rectangular slits with a distance d between adjacent slits. The slits emit light rays coherently and in phase at wavelength λ . The electric field at a distant point P is

$$E_P(r, t) = \sum_{n=0}^{N-1} E_0 \cos(kr - \omega t + n\phi),$$

where E_0 is the amplitude at P of the electric field due to an individual slit, $\phi = 2\pi d \sin \theta / \lambda$, θ is the angle of the rays reaching P (as measured from the perpendicular bisector of the slit arrangement) and r , which is much larger than d , is the distance from P to the most distant slit. Show that the electric field at point P is

$$E_P(r, t) = \left[E_0 \frac{\sin(N\phi/2)}{\sin(\phi/2)} \right] \cos \left[kr - \omega t + (N-2) \frac{\phi}{2} \right]$$

Answer:

$$\begin{aligned} E_P &= \operatorname{Re} \left[\sum_{n=0}^{N-1} E_0 e^{i(kr - \omega t + n\phi)} \right] \\ &= \operatorname{Re} \left[E_0 e^{i(kr - \omega t)} \sum_{n=0}^{N-1} e^{ni\phi} \right] \\ &= \operatorname{Re} \left[E_0 e^{i(kr - \omega t)} \frac{1 - e^{Ni\phi}}{1 - e^{i\phi}} \right] \\ &= \operatorname{Re} \left[E_0 e^{i(kr - \omega t)} \frac{e^{Ni\phi/2} - e^{-Ni\phi/2}}{e^{i\phi/2} - e^{-i\phi/2}} \frac{e^{iN\phi/2}}{e^{i\phi/2}} \right] \\ &= \operatorname{Re} \left[E_0 e^{i(kr - \omega t + (N-1)\phi/2)} \frac{\sin(N\phi/2)}{\sin(\phi/2)} \right] \\ &= \left[E_0 \frac{\sin(N\phi/2)}{\sin(\phi/2)} \right] \cos \left[kr - \omega t + (N-2)\frac{\phi}{2} \right] \end{aligned} \tag{11}$$